Prime harmonics and Floquet topology: A unified operator framework

Branden Lee Friend
Prepared for discussion with Nathan Goldman

Draft: September 16, 2025

Abstract

We develop an operator-theoretic bridge between a prime-harmonic framework and quantized responses in periodically driven (Floquet) quantum matter. The core elements are: (i) a concise recap of the prime-dilation operator and its Fredholm-determinant realization of the Euler product; (ii) a self-contained derivation of Floquet dynamics, quasienergy, and Středa-type response; (iii) an explicit construction of a unified transfer operator whose Abel–Cesàro–summed trace generates a quantized magnetization comparable to a Fredholm determinant identity; and (iv) a bulk–boundary index relation that mirrors a "critical-line stability" mechanism from a spectral-radius argument. We state precise functional hypotheses, summability procedures, and provide a minimal numerical model demonstrating quantized invariants.

Contents

1 Introduction

This manuscript constructs a concrete operator framework linking the Fredholm-determinant / Euler-product machinery from prime-harmonic operators to trace-log regularizations of Floquet response kernels. Our goal is to make the analogy operator-level precise: prime dilations generate an Euler product via a Fredholm determinant; likewise, a Floquet transfer kernel produces an Abel–Cesàro regularized trace-log whose derivative with respect to magnetic flux reproduces an integer-valued winding invariant (Rudner invariant / 3D Chern number), yielding quantized magnetization plateaux.

We present three main theorems: (1) nuclearity and determinant identity for the prime transfer operator in a weighted Hardy–Mellin space; (2) Abel–Cesàro regularization of the Floquet trace-log and its quantized derivative; (3) mapping of a spectral-radius "stability index" in Floquet systems to a critical-manifold picture analogous to the critical line in zeta-function arguments.

The technical hypotheses we require are locality (finite-range or finite Lieb–Robinson velocity), spectral gaps for filtered quasienergy windows, smooth filters g selecting quasienergy arcs, and trace-class (nuclearity) per unit area of the response kernel.

2 Notation and preliminaries

We use complex variable $s = \sigma + \mathrm{i}t, \sigma = \Re s$. Primes are $p \in \mathbb{P}$. Weighted Hardy / Mellin spaces $\mathcal{H}$ will be specified in Section 3. For a trace-class operator A, $\det(I - A)$ denotes the Fredholm determinant and

$$-\log\det(I - A) = \sum_{m=1}^{\infty} \frac{1}{m} \operatorname{Tr}(A^m),$$

convergent in trace-class regime. For divergent series we use Cesàro and Abel summation: for partial sums S_N define

$$\text{Cesáro} \sum_{n\geq 0} a_n = \lim_{N\to\infty} \frac{1}{N+1} \sum_{k=0}^{N} S_k,$$

and Abel (regularization) via

$$\text{Abel} \sum_{n\geq 0} a_n = \lim_{\rho\uparrow 1} \sum_{n\geq 0} a_n \rho^n,$$

when limits exist.

3 Prime-harmonic transfer operator

3.1 Dilation operator and eigenstructure

Define, for each prime p, the dilation operator D_p acting (formally) on functions f on $(0, \infty)$ by

$$(D_p f)(x) = f(px).$$

Generalized eigenfunctions $f_s(x) = x^{-s}$ satisfy

$$D_p f_s = p^{-s} f_s,$$

so multiplicative eigenvalues correspond to p^{-s}.

Interpreting multiplicative eigenvalues as Boltzmann weights e^{-E} gives $E_p = s \log p$ for the prime p.

3.2 Transfer operator and functional setting

Let $\mathcal{H}$ be a weighted Hardy–Mellin Hilbert space with orthonormal basis $\{e_k(x) = x^{-k}\}_{k\geq 0}$ and weights ensuring summability. Concretely one may take coefficients w_k such that the basis becomes orthonormal when equipped with inner product $\langle e_k, e_\ell\rangle = w_k\delta_{k\ell}$; choose w_k decaying fast enough that D_p acts boundedly and desirable operators are nuclear for $\Re s > 1$.

Define the prime-weighted transfer operator

$$T(s) := \sum_p p^{-s} D_p.$$

Acting on basis vectors,

$$D_p e_k = p^{-k} e_k \quad \Rightarrow \quad T(s)e_k = \lambda_k(s)e_k$$

with eigenvalues

$$\lambda_k(s) = \sum_p p^{-(s+k)}.$$

Theorem 3.1 (Prime transfer operator: nuclearity and determinant identity)**.** *Let $\mathcal{H}$ be as above. For $\Re s > 1$ the operator $T(s)$ is trace-class (nuclear) and*

$$\det(I - T(s)) = \prod_p (1 - p^{-s}) = \zeta(s)^{-1}.$$

Proof. We compute the trace norm in the eigenbasis:

$$\|T(s)\|_1 = \sum_{k\geq 0} |\lambda_k(s)| \leq \sum_{k\geq 0}\sum_p p^{-(\sigma+k)} = \sum_p \frac{p^{-\sigma}}{1-p^{-1}}.$$

For $\sigma > 1$, $\sum_p p^{-\sigma} < \infty$ and $\frac{1}{1-p^{-1}} \leq \frac{1}{1-2^{-1}} = 2$, so the double sum is finite and $T(s)$ is trace-class.

The Fredholm expansion holds:

$$-\log\det(I - T) = \sum_{m\geq 1} \frac{1}{m}\operatorname{Tr}(T^m).$$

Evaluate $\operatorname{Tr}(T^m)$ on the basis $\{e_k\}$. For any m-tuple of primes $(p_1, \ldots, p_m)$,

$$D_{p_1}\cdots D_{p_m} = D_{p_1\cdots p_m}$$

acts diagonally with eigenvalue $(p_1\cdots p_m)^{-k}$ on e_k. Summing over $k \geq 0$ yields a geometric denominator factor $\frac{1}{1-(p_1\cdots p_m)^{-1}}$. Collecting and reorganizing terms by prime-power cycles (grouping m-tuples whose product is a given prime power $q = p^r$ and tracking cyclic equivalence classes) reduces the trace-log series to

$$\sum_{m\geq 1}\frac{1}{m}\operatorname{Tr}(T^m) = \sum_p\sum_{r\geq 1}\frac{1}{r}p^{-rs} = -\sum_p \log(1-p^{-s}).$$

Exponentiating gives the Euler product identity $\det(I - T) = \prod_p(1 - p^{-s}) = \zeta(s)^{-1}$. □

Remark 3.2. The combinatorial rearrangement step is standard in transfer-operator treatments of Euler products; Appendix A gives the explicit grouping by prime-power cycles and justifies interchange of sums in the nuclear regime $\Re s > 1$.

4 Floquet systems and Středa-type responses

4.1 Floquet evolution and quasienergy

Let $H(t)$ be time-periodic with period T. The time-ordered propagator is

$$U(t,0) = \mathcal{T} \exp\Big(-\frac{\mathrm{i}}{\hbar} \int_0^t H(\tau)\,\mathrm{d}\tau \Big),$$

and the one-period Floquet operator is $U_F := U(T,0)$. Quasienergies ε_α are defined modulo $\hbar\Omega$ with $\Omega = 2\pi/T$ by

$$U_F|\phi_\alpha\rangle = \mathrm{e}^{-\mathrm{i}\varepsilon_\alpha T/\hbar}|\phi_\alpha\rangle.$$

4.2 Středa formula (static baseline)

For static 2D systems, at zero temperature, the Středa relation relates density and magnetic field:

$$\frac{\partial n}{\partial B}\Big|_\mu = \frac{e}{h} C,$$

where C is the Chern number of filled bands. This follows from the thermodynamic identity $\partial_B n|_\mu = \partial_\mu M|_B$ combined with Kubo formula linking Hall conductivity to Berry curvature and hence Chern number.

4.3 Floquet generalization and energy-resolved response

For periodic driving define a quasienergy-resolved density of states $\rho(\varepsilon, B)$ on the Floquet Brillouin zone (circle) and a distribution $f(\varepsilon)$ describing occupation. A generalized Středa relation can be written formally as

$$\frac{\partial n}{\partial B} = \int_{\mathrm{FBZ}} \frac{\mathrm{d}\varepsilon}{2\pi}\, f(\varepsilon)\, \mathcal{C}(\varepsilon),$$

where $\mathcal{C}(\varepsilon)$
is a Floquet-Berry curvature/winding density. In practice, summations over Floquet replicas lead to conditionally convergent series and require summability regularization (Cesàro/Abel).

5 Unified Floquet–magnetic transfer operator and trace-log

5.1 Definition of the Floquet response kernel

Let $U_F(B)$ be the Floquet operator in presence of a weak uniform magnetic field B. Choose a smooth, periodic filter g selecting a quasienergy arc and define

$$P_g(B) := g\Big(\frac{\mathrm{i}}{T} \log U_F(B)\Big),$$

where the branch of log is fixed consistently with the chosen Floquet zone. Let $\mathcal{V}$ be a locality-preserving response vertex (e.g., current–current kernel), and α a normalization constant. Define the Floquet transfer kernel

$$\mathcal{T}(B) := \alpha\, P_g(B)\, \mathcal{V}\, P_g(B).$$

Working on the single-particle Hilbert space and then taking trace-per-area in the thermodynamic limit yields a trace-class property under locality and spectral-gap hypotheses.

5.2 Generating functional and regularization

Define the (formal) generating functional

$$\Phi(\lambda; B) := -\log\det\big(I - \lambda\,\mathcal{T}(B)\big) = \sum_{m\geq 1} \frac{\lambda^m}{m}\,\mathrm{Tr}_{\square}\big(\mathcal{T}(B)^m\big).$$

Because the Floquet spectrum is a circle, naive replica sums over Floquet zones can produce conditionally convergent series. Introduce an Abel regulator $\rho \in (0,1)$ and set

$$\Phi_\rho(1; B) := \sum_{m\geq 1} \frac{\rho^m}{m}\,\mathrm{Tr}_{\square}(\mathcal{T}(B)^m),$$

then define the physically meaningful value as

$$\Phi(1; B) := \lim_{\rho\uparrow 1} \Phi_\rho(1; B),$$

provided the limit exists and is independent of the Abel path; under Tauberian hypotheses this equals the Cesàro limit.

Theorem 5.1 (Floquet transfer kernel and Abel–Cesàro quantization)**.** *Under locality, spectral-gap, and smooth-filter hypotheses, the Floquet transfer kernel $\mathcal{T}(B)$ is trace-class per unit area and the Abel-regularized functional $\Phi_\rho(1; B)$ admits a limit $\Phi(1; B) = \lim_{\rho\uparrow 1}\Phi_\rho(1; B)$. Moreover,*

$$M(B) := \partial_B \Phi(1; B) = \frac{e}{h}\,\mathcal{W}[U_F; g],$$

where $\mathcal{W}$ is an integer-valued winding invariant (Rudner winding or 3D Chern number on (k_x, k_y, t)).

Proof (outline). Each $\mathrm{Tr}_{\square}(\mathcal{T}^m)$
can be expressed as an m-fold
time-ordered correlation integral over one period which, in the adiabatic/gapped regime, reduces to an integral of a winding-density on the 3-torus (k_x, k_y, t).

Locality (finite Lieb–Robinson velocity) ensures subexponential growth bounds on $\mathrm{Tr}(\mathcal{T}^m)$ permitting Abel summation.

Under Tauberian conditions (Hardy–Littlewood type) Abel and Cesàro sums coincide for such bounded growth. Homotopy invariance of the determinant resummation implies that the derivative with respect to B is integer-quantized and equals a topological winding $\mathcal{W}$.

for Tauberian details and Appendix for the reduction of correlation integrals to winding densities.

□

6 Stability index and critical manifold correspondence

Define the stability index

$$\beta(B) := \log r(\mathcal{T}(B)),$$

where $r(\cdot)$ denotes the spectral radius. In topological phases protected by a gap the spectral properties of $\mathcal{T}(B)$ are robust under smooth deformations and $\beta(B)$ remains pinned except when the gap closes.

Theorem 6.1 (Stability index mapping)**.** *Under the previous hypotheses, $\beta(B)$ is topologically pinned within phases and changes sign only at topological transitions (gap closings). The set $\{\beta = 0\}$ is the Floquet analogue of a critical manifold similar in role to the critical line $\Re s = \frac{1}{2}$ in prime-transfer spectral-radius arguments.*

Sketch. The largest singular value controls iteration growth; topology fixes invariants (winding numbers) that prevent continuous crossing of spectral radius thresholds without a closing of the spectral gap. Thus sign change events in β coincide with phase transitions where winding invariants jump. □

7 Minimal numerical model and results

To produce a concrete demonstration, we consider a minimal two-step driven two-band lattice model (Qi–Wu–Zhang style). Define a Bloch Hamiltonian vector $\boldsymbol{d}(\boldsymbol{k};t)$ for two-step drive with parameters m_1, m_2 and step durations T_1, T_2. The one-period Floquet operator is $U_F(\boldsymbol{k}) = U_2(\boldsymbol{k})U_1(\boldsymbol{k})$ with $U_j(\boldsymbol{k}) = \exp\big(-\mathrm{i}\,\boldsymbol{d}_j(\boldsymbol{k})\cdot\boldsymbol{\sigma}\,T_j\big)$. We compute eigenphases and eigenvectors of $U_F(\boldsymbol{k})$ on a 41×41 grid on the Brillouin zone and evaluate the Chern number of a chosen quasienergy band using the Fukui discretization (plaquette phase product).

7.1 Model and parameter sweep

The particular d-vector used in numerics is

$$\boldsymbol{d}(\boldsymbol{k};m) = (A\sin k_x,\ A\sin k_y,\ m + \cos k_x + \cos k_y),$$

with $A = 1$ and $T_1 = T_2 = 1$. We sweep $m_2 \in [-2.5, 2.5]$ with $m_1 = 0.5$ fixed.

7.2 Numerical results (table)

The computed Chern numbers for the principal Floquet band at each m_2 are:

m_2	Chern
-2.500	2
-2.250	2
-2.000	2
-1.750	2
-1.500	2
-1.250	1
-1.000	1
-0.750	1
-0.500	1
-0.250	-1
0.000	-1
0.250	-1
0.500	-2
0.750	-2
1.000	-2
1.250	-2
1.500	-2
1.750	-2
2.000	-1
2.250	-1
2.500	-1

(The sweep points are 21 evenly spaced values from -2.5 to 2.5.)

Schematic operator-level mapping: primes $\leftrightarrow$ Floquet transfer kernels; Euler product matches determinant identity; winding invariant corresponds to quantized magnetization.

Minimal model: Floquet Chern number vs mass parameter m_2.

8 Consequences, predictions and experimental notes

The operator framework predicts:

- For gapped driven 2D systems with a chosen quasienergy cut, the magnetization per area computed from the Abel–Cesàro-regularized trace-log is quantized to $\frac{e}{h}\mathcal{W}$ and exhibits plateaux as B varies until a topological transition.
- Energy pumping into (or from) the drive has a Středa-type back-action: average absorbed power is the frequency times an integral of the curvature density.
- Disorder robustness: noncommutative geometry / trace-per-volume methods replace Bloch integrals; quantization persists until mobility gaps close.
- Stability index $\beta = \log r(\mathcal{T})$ changes sign only at topological transitions; the locus $\beta = 0$ plays the analogue role of a critical manifold.

9 Limitations and open problems

- **Rigorous class of models:** Trace-class property of $\mathcal{T}$ depends on locality and the specific filter g. Interacting Floquet systems are substantially more delicate.
- **Branch choices:** The operator $\log U_F$ requires branch choice; physical observables are consistent up to integer shifts, but a careful treatment must track branch dependence.
- **Number-theoretic equivalence:** The correspondence is structural and operator-level; we do not claim that Floquet physics implies new statements about zeros of $\zeta(s)$.

A Detailed derivations: nuclearity and Fredholm expansion

A.1 N

uclearity bound for $T(s)$

For $\sigma = \Re s > 1$ we bound

$$\|T(s)\|_1 = \sum_{k\geq 0}\left|\sum_p p^{-(s+k)}\right| \leq \sum_{k\geq 0}\sum_p p^{-(\sigma+k)} = \sum_p \frac{p^{-\sigma}}{1-p^{-1}}.$$

Using the elementary estimate $\frac{1}{1-p^{-1}} \leq \frac{1}{1-2^{-1}} = 2$, we obtain

$$\|T(s)\|_1 \leq 2\sum_p p^{-\sigma} < \infty,$$

since $\sum_p p^{-\sigma}$ converges for $\sigma > 1$. Thus $T(s)$ is nuclear (trace-class).

A.2 A2. Fredholm determinant combinatorics

Write

$$\mathrm{Tr}(T^m) = \sum_{k\geq 0}\langle e_k, T^m e_k\rangle = \sum_{k\geq 0}\sum_{(p_1,\ldots,p_m)}(p_1\cdots p_m)^{-(s+k)}.$$

Swap sums (justified by absolute convergence for $\sigma > 1$):

$$\mathrm{Tr}(T^m) = \sum_{(p_1,\ldots,p_m)}(p_1\cdots p_m)^{-s}\sum_{k\geq 0}(p_1\cdots p_m)^{-k} = \sum_{(p_1,\ldots,p_m)}\frac{(p_1\cdots p_m)^{-s}}{1-(p_1\cdots p_m)^{-1}}.$$

Now group m-tuples by their multiplicative product $q = p^r$. The factor $1/[1-q^{-1}]$ cancels partial geometric overcounting when restricting to cyclic equivalence classes (cycle index computations). The standard prime-power cycle enumeration yields

$$\sum_{m\geq 1}\frac{1}{m}\,\mathrm{Tr}(T^m) = \sum_p\sum_{r\geq 1}\frac{1}{r}p^{-rs} = -\sum_p \log(1-p^{-s}).$$

Exponentiating completes the identity.

B Tauberian control and Abel–Cesàro equivalence

B.1 A3. Abel vs Cesàro for subexponential growth

Let $a_m = \frac{1}{m}\,\mathrm{Tr}(\mathcal{T}^m)$. Suppose there exist constants $C, \gamma > 0$ such that $a_m \le C\mathrm{e}^{\gamma m}$ with γ sufficiently small (Tauberian control). Then the Abel sum

$$A(\rho) := \sum_{m\ge 1} \rho^m a_m$$

is analytic for $0 \le \rho < \mathrm{e}^{-\gamma}$ and admits a limit as $\rho \uparrow 1$ under standard Tauberian hypotheses. The classical Hardy–Littlewood Tauberian theorem gives conditions under which

$$\lim_{\rho\uparrow 1} A(\rho) = \text{Cesáro} \sum_{m\ge 1} a_m.$$

In our context locality and finite propagation speed (Lieb–Robinson) imply bounds on correlation function spread and hence on growth of $\mathrm{Tr}(\mathcal{T}^m)$, enabling Tauberian conclusions.

B.2 A4. Středa baseline (static)

Standard derivation: thermodynamic potential $\Omega(\mu, B, T)$, density $n = -\partial_\mu \Omega$, magnetization $M = -\partial_B \Omega$. Maxwell relation yields

$$\frac{\partial n}{\partial B}\Big|_\mu = \frac{\partial M}{\partial \mu}\Big|_B.$$

For an insulator with Fermi level in a gap, the Kubo formula computes $\partial_\mu M = (e/h)C$, proving the Středa relation.

C Appendix B: Minimal-model code (Python) — reproducible numerics

This is the code used to create the numerical figure (Chern sweep) and to compute the table above.

```
#!/usr/bin/env python3
# floquet_model.py -- minimal two-band Floquet model and Chern-number computation
# Usage: python3 floquet_model.py

import numpy as np

# Pauli matrices
sigma_x = np.array([[0,1],[1,0]], dtype=complex)
sigma_y = np.array([[0,-1j],[1j,0]], dtype=complex)
sigma_z = np.array([[1,0],[0,-1]], dtype=complex)
identity = np.eye(2, dtype=complex)

def U_from_d(dvec, t=1.0):
    """Compute exp(-i (d·) t) using analytic 2x2 formula."""
```

```
    d = np.array(dvec, dtype=float)
    norm = np.linalg.norm(d)
    if norm < 1e-12:
        return identity.copy()
    dhat = d / norm
    c = np.cos(norm * t)
    s = np.sin(norm * t)
    dhat_sigma = dhat[0]*sigma_x + dhat[1]*sigma_y + dhat[2]*sigma_z
    return c*identity - 1j*s*dhat_sigma

def H_qwz_d(kx, ky, m, A=1.0):
    """d-vector for Qi--Wu--Zhang style Hamiltonian"""
    return np.array([A*np.sin(kx), A*np.sin(ky), m + np.cos(kx) + np.cos(ky)])

def compute_floquet_U(kx, ky, m1, m2, T1=1.0, T2=1.0, A=1.0):
    U1 = U_from_d(H_qwz_d(kx, ky, m1, A), t=T1)
    U2 = U_from_d(H_qwz_d(kx, ky, m2, A), t=T2)
    return U2 @ U1

def floquet_band_eig(U):
    vals, vecs = np.linalg.eig(U)
    phases = np.angle(vals)
    idx = np.argsort(phases)
    return phases[idx], vecs[:, idx]

def compute_chern_on_grid(m1, m2, Nk=41):
    kxs = np.linspace(-np.pi, np.pi, Nk, endpoint=False)
    kys = np.linspace(-np.pi, np.pi, Nk, endpoint=False)
    band_vec = np.zeros((Nk, Nk, 2), dtype=complex)
    for i, kx in enumerate(kxs):
        for j, ky in enumerate(kys):
            U = compute_floquet_U(kx, ky, m1, m2, T1=1.0, T2=1.0)
            phases, vecs = floquet_band_eig(U)
            v = vecs[:,0]
            band_vec[i,j,:] = v / np.linalg.norm(v)
    # Fukui plaquette product
    chern_sum = 0.0
    for i in range(Nk):
        for j in range(Nk):
            ip = (i+1) % Nk
            jp = (j+1) % Nk
            u = band_vec[i,j]
            ux = band_vec[ip,j]
            uy = band_vec[i,jp]
            ux_y = band_vec[ip,jp]
            U1 = np.vdot(u, ux)
            U2 = np.vdot(ux, ux_y)
            U3 = np.vdot(uy, ux_y)
```

```
            U4 = np.vdot(u, uy)
            W = U1 * U2 * np.conj(U3) * np.conj(U4)
            chern_sum += np.angle(W)
    chern = chern_sum / (2*np.pi)
    return int(np.round(chern))

if __name__ == "__main__":
    m1 = 0.5
    m2_values = np.linspace(-2.5, 2.5, 21)
    chern_values = []
    for m2 in m2_values:
        ch = compute_chern_on_grid(m1, float(m2), Nk=41)
        chern_values.append(ch)
        print(f"m2 = {m2:6.3f}  ->  Chern = {ch}")
    # Optionally save to CSV
    import csv
    with open("chern_sweep.csv","w",newline="") as fh:
        writer = csv.writer(fh)
        writer.writerow(["m2","chern"])
        for m2, ch in zip(m2_values, chern_values):
            writer.writerow([m2, ch])
    print("Done. Data saved to chern_sweep.csv")

\end{verbatim

\section{Appendix C: Figure-generation scripts}

If you prefer to generate the figures yourself, here are the scripts I used.

\subsection*{Script A: schematic figure (matplotlib)}
Save as ‘fig_schematic.py‘ (or run in a notebook):
\begin{verbatim}
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(8,4.5), dpi=200)
ax.axis(’off’)
ax.set_title("Schematic: Prime-harmonics  Floquet topology", fontsize=14, fontweight=’bold’)
ax.text(0.05, 0.75, "Prime dilation operator\n$D_p:\\;f(x)\\mapsto f(px)$", fontsize=10, bbox=d
ax.text(0.35, 0.75, "Transfer operator\n$T(s)=\\sum_p p^{-s} D_p$\nFredholm determinant", fonts
ax.arrow(0.22, 0.73, 0.08, 0, head_width=0.03, head_length=0.02)
ax.text(0.05, 0.35, r"Floquet operator $U_F=\mathcal{T}e^{-\frac{i}{\hbar}\int_0^T H(t)\,dt}$"
ax.text(0.35, 0.35, r"Floquet transfer kernel $\mathcal{T}=P_g\,\mathcal{V}\,P_g$\nTrace-log &
ax.arrow(0.22, 0.33, 0.08, 0, head_width=0.03, head_length=0.02)
ax.annotate("", xy=(0.45,0.7), xytext=(0.45,0.5), arrowprops=dict(arrowstyle="->"))
ax.text(0.47, 0.58, "Operator-level\nanalogy", fontsize=10)
ax.text(0.78, 0.75, r"Euler product $\prod_p (1-p^{-s})^{-1}$", fontsize=10)
ax.text(0.78, 0.35, r"Rudner winding / 3D Chern number $\mathcal{W}$", fontsize=10)
fig.savefig("fig_schematic.png", bbox_inches=’tight’)
\end{verbatim
```

```
\subsection*{Script B: Chern sweep plot}
(Use the 'floquet_model.py' above to compute CSV, then plot)
\begin{verbatim}
import numpy as np, matplotlib.pyplot as plt, csv

m2 = []
chern = []
with open("chern_sweep.csv","r") as fh:
    r = csv.reader(fh)
    next(r)
    for row in r:
        m2.append(float(row[0])); chern.append(int(row[1]))

plt.figure(figsize=(7,4))
plt.plot(m2, chern, marker='o', linestyle='-')
plt.xlabel("Drive mass parameter $m_2$ (with $m_1=0.5$)")
plt.ylabel("Computed Chern number (Floquet band)")
plt.title("Minimal model: Floquet Chern number vs $m_2$")
plt.grid(True)
plt.savefig("fig_chern_sweep.png", bbox_inches='tight', dpi=300)
\end{verbatim

#!/usr/bin/env python3
# floquet_model.py -- minimal two-band Floquet model and Chern-number computation
# Usage: python3 floquet_model.py

import numpy as np

# Pauli matrices
sigma_x = np.array([[0,1],[1,0]], dtype=complex)
sigma_y = np.array([[0,-1j],[1j,0]], dtype=complex)
sigma_z = np.array([[1,0],[0,-1]], dtype=complex)
identity = np.eye(2, dtype=complex)

def U_from_d(dvec, t=1.0):
    """Compute exp(-i (d·) t) using analytic 2x2 formula."""
    d = np.array(dvec, dtype=float)
    norm = np.linalg.norm(d)
    if norm < 1e-12:
        return identity.copy()
    dhat = d / norm
    c = np.cos(norm * t)
    s = np.sin(norm * t)
    dhat_sigma = dhat[0]*sigma_x + dhat[1]*sigma_y + dhat[2]*sigma_z
    return c*identity - 1j*s*dhat_sigma

def H_qwz_d(kx, ky, m, A=1.0):
```

```python
    """d-vector for Qi--Wu--Zhang style Hamiltonian"""
    return np.array([A*np.sin(kx), A*np.sin(ky), m + np.cos(kx) + np.cos(ky)])

def compute_floquet_U(kx, ky, m1, m2, T1=1.0, T2=1.0, A=1.0):
    U1 = U_from_d(H_qwz_d(kx, ky, m1, A), t=T1)
    U2 = U_from_d(H_qwz_d(kx, ky, m2, A), t=T2)
    return U2 @ U1

def floquet_band_eig(U):
    vals, vecs = np.linalg.eig(U)
    phases = np.angle(vals)
    idx = np.argsort(phases)
    return phases[idx], vecs[:, idx]

def compute_chern_on_grid(m1, m2, Nk=41):
    kxs = np.linspace(-np.pi, np.pi, Nk, endpoint=False)
    kys = np.linspace(-np.pi, np.pi, Nk, endpoint=False)
    band_vec = np.zeros((Nk, Nk, 2), dtype=complex)
    for i, kx in enumerate(kxs):
        for j, ky in enumerate(kys):
            U = compute_floquet_U(kx, ky, m1, m2, T1=1.0, T2=1.0)
            phases, vecs = floquet_band_eig(U)
            v = vecs[:,0]
            band_vec[i,j,:] = v / np.linalg.norm(v)
    # Fukui plaquette product
    chern_sum = 0.0
    for i in range(Nk):
        for j in range(Nk):
            ip = (i+1) % Nk
            jp = (j+1) % Nk
            u = band_vec[i,j]
            ux = band_vec[ip,j]
            uy = band_vec[i,jp]
            ux_y = band_vec[ip,jp]
            U1 = np.vdot(u, ux)
            U2 = np.vdot(ux, ux_y)
            U3 = np.vdot(uy, ux_y)
            U4 = np.vdot(u, uy)
            W = U1 * U2 * np.conj(U3) * np.conj(U4)
            chern_sum += np.angle(W)
    chern = chern_sum / (2*np.pi)
    return int(np.round(chern))

if __name__ == "__main__":
    m1 = 0.5
    m2_values = np.linspace(-2.5, 2.5, 21)
    chern_values = []
    for m2 in m2_values:
```

```
        ch = compute_chern_on_grid(m1, float(m2), Nk=41)
        chern_values.append(ch)
        print(f"m2 = {m2:6.3f}  ->  Chern = {ch}")
    # Optionally save to CSV
    import csv
    with open("chern_sweep.csv","w",newline="") as fh:
        writer = csv.writer(fh)
        writer.writerow(["m2","chern"])
        for m2, ch in zip(m2_values, chern_values):
            writer.writerow([m2, ch])
    print("Done. Data saved to chern_sweep.csv")

import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(8,4.5), dpi=200)
ax.axis('off')
ax.set_title("Schematic: Prime-harmonics  Floquet topology", fontsize=14, fontweight='bold')
ax.text(0.05, 0.75, "Prime dilation operator\n$D_p:\\;f(x)\\mapsto f(px)$", fontsize=10, bbox=d
ax.text(0.35, 0.75, "Transfer operator\n$T(s)=\\sum_p p^{-s} D_p$\nFredholm determinant", fonts
ax.arrow(0.22, 0.73, 0.08, 0, head_width=0.03, head_length=0.02)
ax.text(0.05, 0.35, r"Floquet operator $U_F=\mathcal{T}e^{-\frac{i}{\hbar}\int_0^T H(t)\,dt}$"
ax.text(0.35, 0.35, r"Floquet transfer kernel $\mathcal{T}=P_g\,\mathcal{V}\,P_g$\nTrace-log &
ax.arrow(0.22, 0.33, 0.08, 0, head_width=0.03, head_length=0.02)
ax.annotate("", xy=(0.45,0.7), xytext=(0.45,0.5), arrowprops=dict(arrowstyle="->"))
ax.text(0.47, 0.58, "Operator-level\nanalogy", fontsize=10)
ax.text(0.78, 0.75, r"Euler product $\prod_p (1-p^{-s})^{-1}$", fontsize=10)
ax.text(0.78, 0.35, r"Rudner winding / 3D Chern number $\mathcal{W}$", fontsize=10)
fig.savefig("fig_schematic.png", bbox_inches='tight')

import numpy as np, matplotlib.pyplot as plt, csv

m2 = []
chern = []
with open("chern_sweep.csv","r") as fh:
    r = csv.reader(fh)
    next(r)
    for row in r:
        m2.append(float(row[0])); chern.append(int(row[1]))

plt.figure(figsize=(7,4))
plt.plot(m2, chern, marker='o', linestyle='-')
plt.xlabel("Drive mass parameter $m_2$ (with $m_1=0.5$)")
plt.ylabel("Computed Chern number (Floquet band)")
plt.title("Minimal model: Floquet Chern number vs $m_2$")
plt.grid(True)
plt.savefig("fig_chern_sweep.png", bbox_inches='tight', dpi=300)
```

```
m2 values (21 points) and computed Chern numbers:
-2.500 ->  2
```

```
-2.250 ->  2
-2.000 ->  2
-1.750 ->  2
-1.500 ->  2
-1.250 ->  1
-1.000 ->  1
-0.750 ->  1
-0.500 ->  1
-0.250 -> -1
 0.000 -> -1
 0.250 -> -1
 0.500 -> -2
 0.750 -> -2
 1.000 -> -2
 1.250 -> -2
 1.500 -> -2
 1.750 -> -2
 2.000 -> -1
 2.250 -> -1
 2.500 -> -1
```

D Acknowledgements

Author: Branden Lee Friend. Contact: brandenfriend007@outlook.com